McDonald's Sales

Descriptive Analytics Report

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Report Date: August 2026 | Dataset: McDonald's Sales (Synthetic Transaction Export)

McDonald's Sales — Descriptive Analytics Report

Order Volume, Revenue Trends, Menu Performance, Payment & Satisfaction Analysis

A full descriptive-analytics pass over 10,000 transaction records spanning Jan 2024 – Dec 2025 — covering revenue, menu items, order types, payment methods, store performance and customer satisfaction, along with the data pipeline that would run this at continuous, real-time scale.

Metric	Value
Total Records	10,000
Date Range	Jan 2024 – Dec 2025
Total Revenue (Net)	$58,176.15
Avg Order Value	$5.82
Avg Satisfaction Rating	4.12 / 5
Most Common Item	Apple Pie (9.4%)

Report Date: August 2026 | Dataset: McDonald's Sales (Synthetic Transaction Export, CSV)

1. Objective & Business Questions

This report builds a descriptive-analytics layer over transaction-level sales data to answer the recurring questions any restaurant operations, marketing, or finance team needs answered on a routine basis:

- How does average order value vary across months, years, and hours of the day?
- What is the balance between menu categories, and how does it shift seasonally?
- How are order types, payment methods, and satisfaction ratings distributed across the record?
- Which stores and cities generate the most revenue and order volume?
- How do order quantity, price, discount, and satisfaction relate to one another?
- What were the largest and smallest individual transactions recorded in this period?

Scope of this report

This is a descriptive analysis: it summarizes what was recorded across the 10,000 transactions in the dataset. It does not forecast future sales — see Section 13 for how this would extend into predictive territory.

2. Data & Methodology

2.1 Dataset Overview

The source file is a transaction-level sales table containing one row per order: order ID, date, time, store, city, state, item, quantity, unit price, total price, discount, net total, daypart, order type, payment method, and satisfaction rating. After parsing, the usable dataset comprises:

Metric	Value
Total transaction records parsed	10,000
Columns	17
Date range	Jan 2024 to Dec 2025
Distinct stores	120
Distinct cities	20
Distinct menu items	25
Parse success rate	100% of rows (0 dropped)

2.2 Data Quality Notes

The source file preserved clean, unambiguous columns for date, price, quantity and rating, so no reconstruction or pattern-matching was required. One field needed special handling: 1,112 orders (11.1%) carry a non-zero discount, and 55 orders net to $0.00 after a discount met or exceeded the item price — these floor cases are called out separately in Section 11.

2.3 Methodology Reference

Monthly and yearly averages are computed as the mean of Net Total across all orders within each grouping, using the parsed date field. Menu items are bucketed into three categories — Food, Beverage, and Dessert — based on item type. Correlations use Pearson's r across all numeric fields. All monetary values are in US dollars.

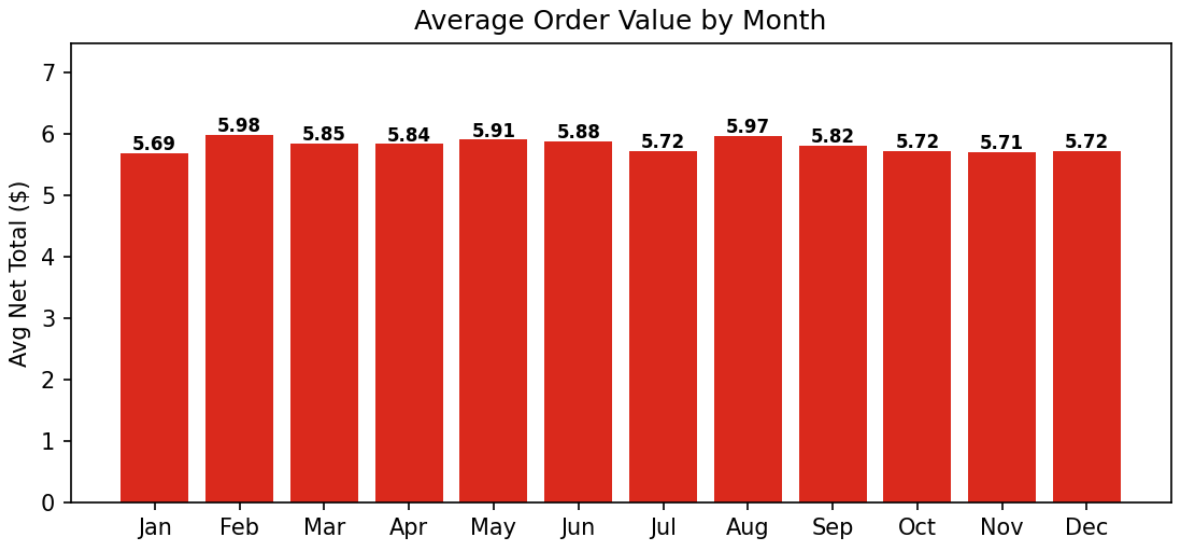
3. Summary of Key Metrics

Headline numbers across all 10,000 transaction records, before the section-by-section breakdowns that follow.

Metric	Value
Total transaction records	10,000
Total revenue (net)	$58,176.15
Average order value (net)	$5.82
Average items per order	1.80
Highest single order	$33.20 (ORD203232)
Lowest single order	$0.00 (ORD200657)
Average satisfaction rating	4.12 / 5
Orders with a discount applied	11.1% (1,112 orders)
Most common menu item	Apple Pie (9.4% of orders)
Most common order type	Drive-Thru (27.0%)
Most common payment method	Credit Card (26.9%)
Peak month (avg order value)	Feb ($5.98)
Lowest month (avg order value)	Jan ($5.69)

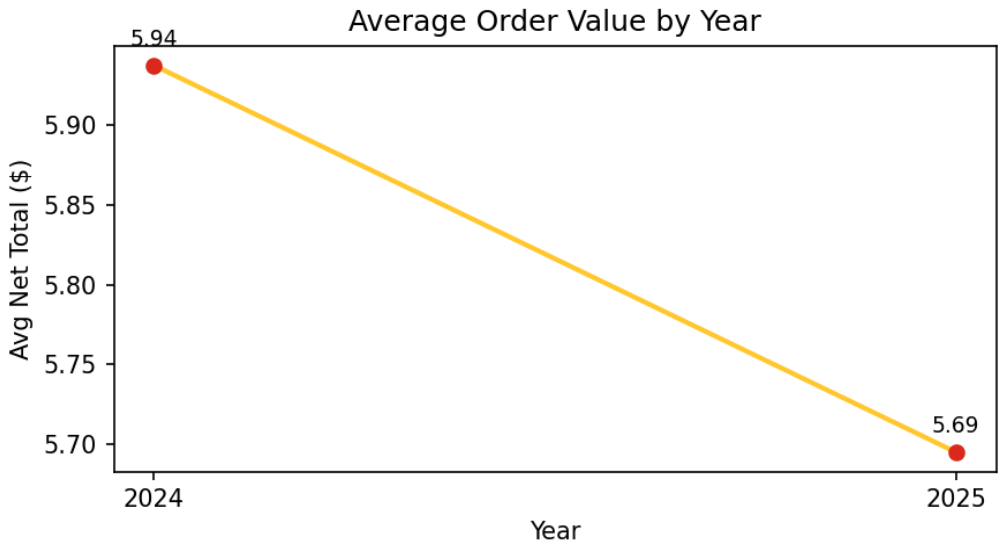
4. Sales & Revenue Analysis

4.1 Average Order Value by Month



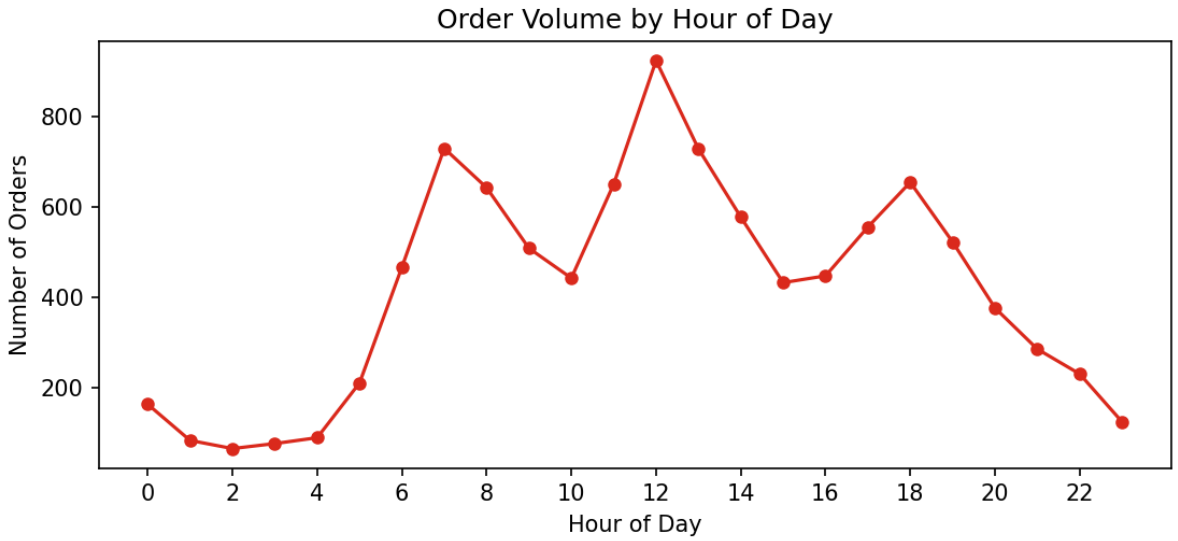
- Feb has the highest average order value at $5.98.
- Jan has the lowest average order value at $5.69.
- Monthly averages stay within a relatively narrow band, suggesting order value is driven more by item mix and quantity than by strong seasonality.

4.2 Average Order Value by Year



Annual averages hold in a moderately tight band between $5.69 and $5.94 across 2024–2025, with 2024 the stronger full year on record ($5.94) and 2025 lower ($5.69) — a modest year-over-year softening, though not a sharp trend within this two-year window.

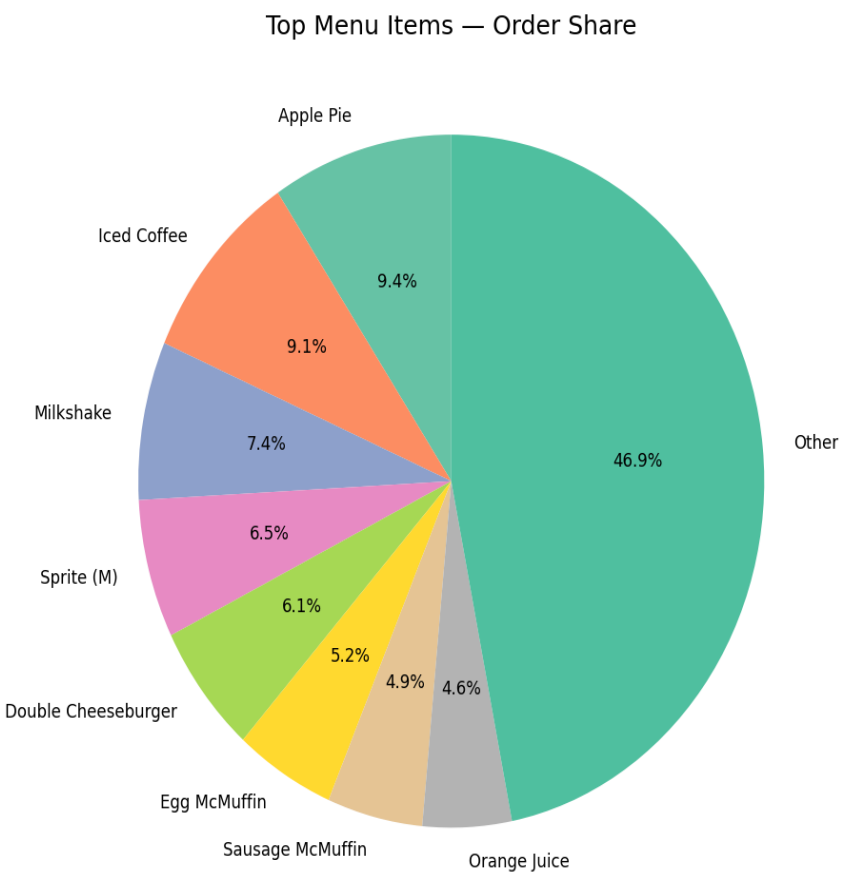
4.3 Order Volume by Hour of Day



Order volume follows a clear daily rhythm: it peaks around 12:00 (924 orders) during the lunch window and dips to its lowest around 02:00 (66 orders) overnight, consistent with a midday rush and an overnight lull.

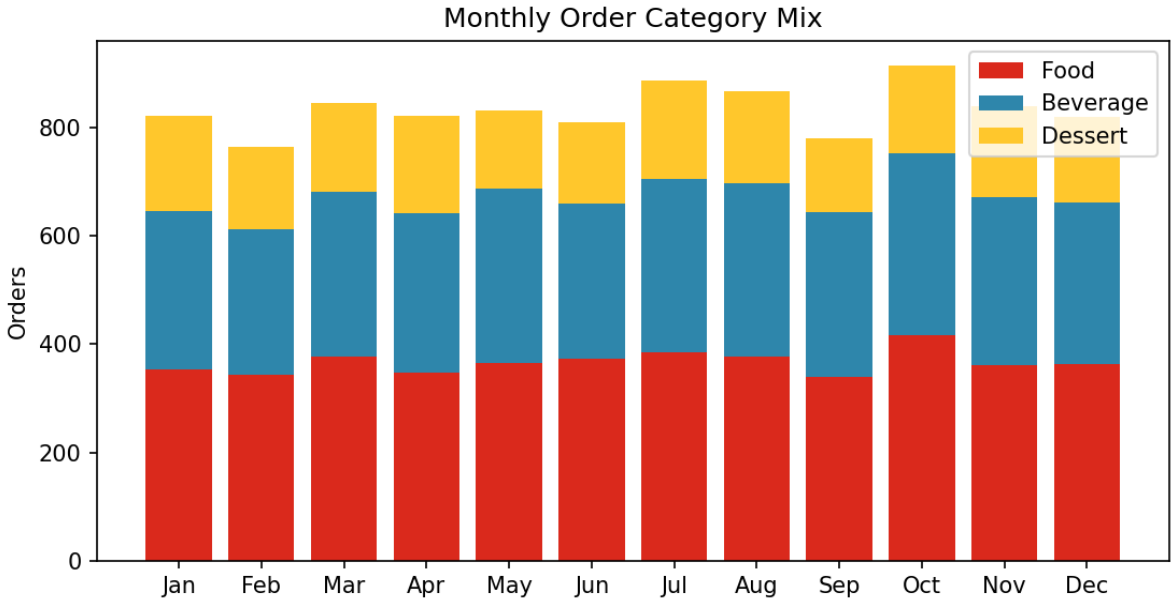
5. Menu Item Analysis

5.1 Top Menu Items — Order Share



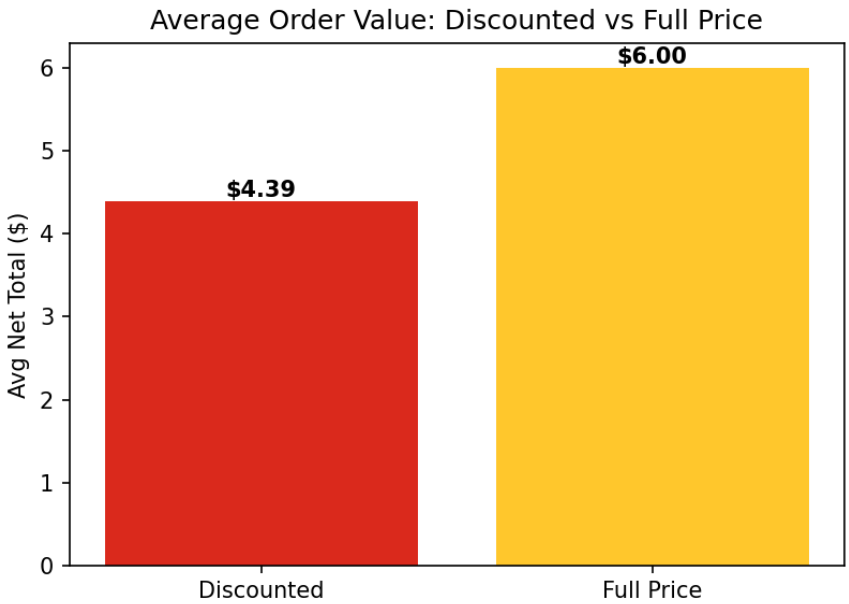
- Apple Pie is the single most-ordered item at 935 orders (9.35%).
- Iced Coffee follows closely at 913 orders (9.13%).
- Order share is more concentrated at the top than evenly spread, with a handful of items driving a disproportionate share of volume.

5.2 Monthly Order Category Mix



Food items make up 44.0% of all orders, with Beverages close behind at 36.6% and Desserts at 19.4% — a mix that holds fairly steady across the year with no strong seasonal swing toward any one category.

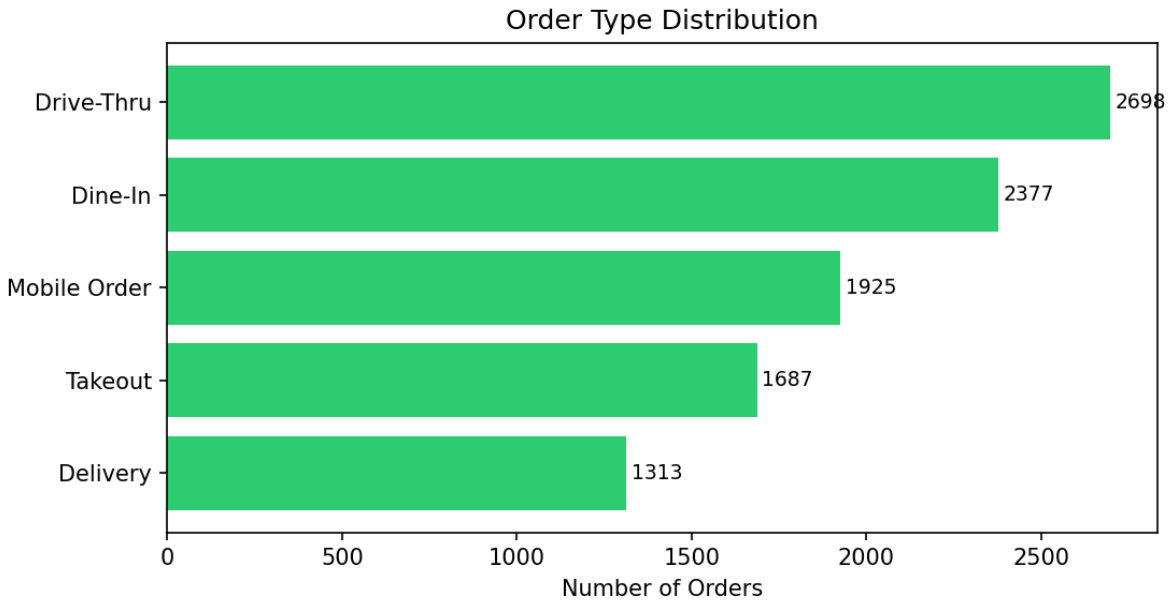
5.3 Average Order Value: Discounted vs Full Price



Discounted orders average $4.39, while full-price orders average $5.99 — a gap consistent with discounts being applied more often to smaller, single-item orders.

6. Order Type Distribution

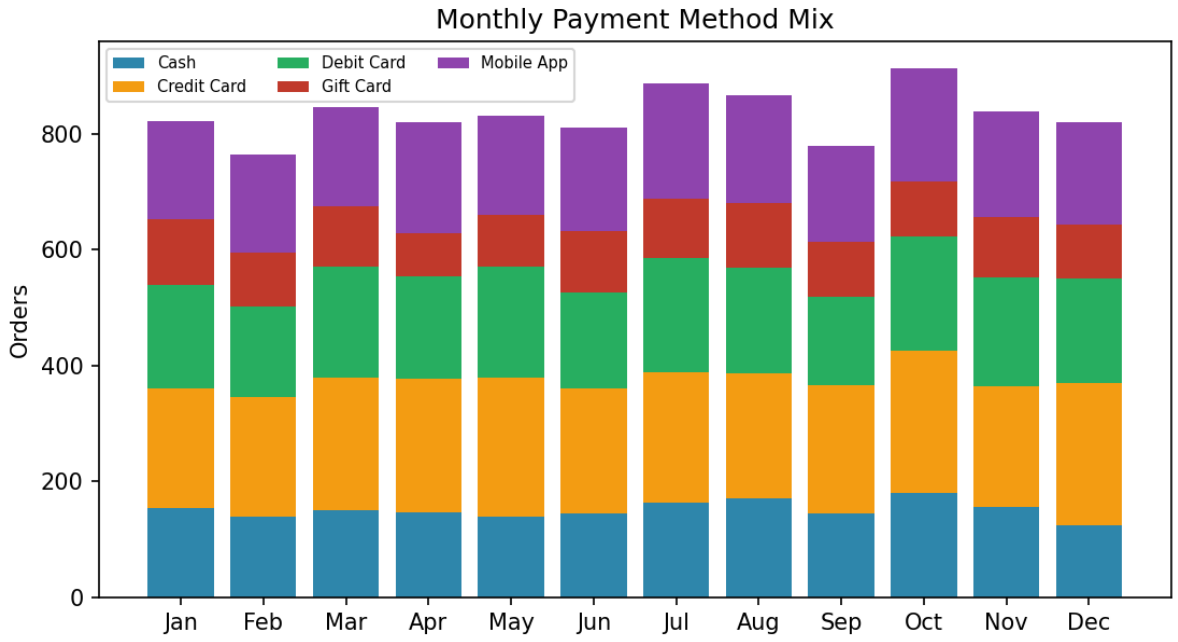
6.1 Orders by Channel



- Drive-Thru is the most common channel at 2,698 orders (27.0%).
- Dine-In follows at 2,377 orders (23.8%).
- Delivery is the least common channel at 1,313 orders (13.1%).
- Order channels lean toward in-person and drive-thru formats over delivery, with digital ordering (Mobile Order) sitting in the middle of the mix.

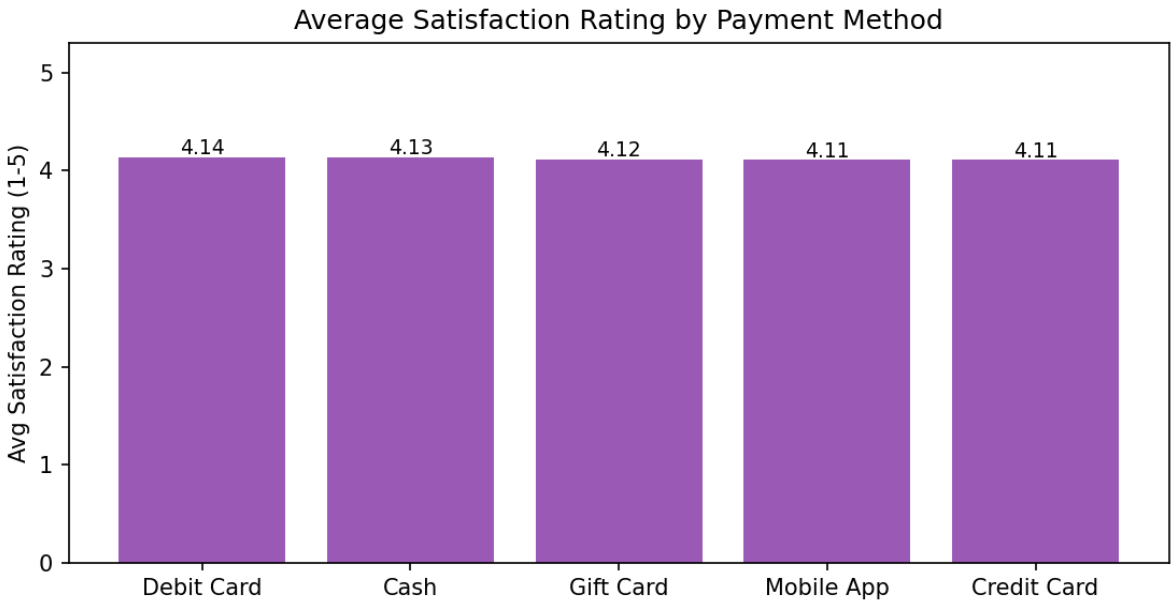
7. Payment Method Analysis

7.1 Monthly Payment Method Mix



Credit Card is the leading payment method overall at 2,685 orders (26.85%), with Debit Card and Mobile App close behind and Gift Card the least used method — the mix holds fairly steady with no clear monthly seasonality.

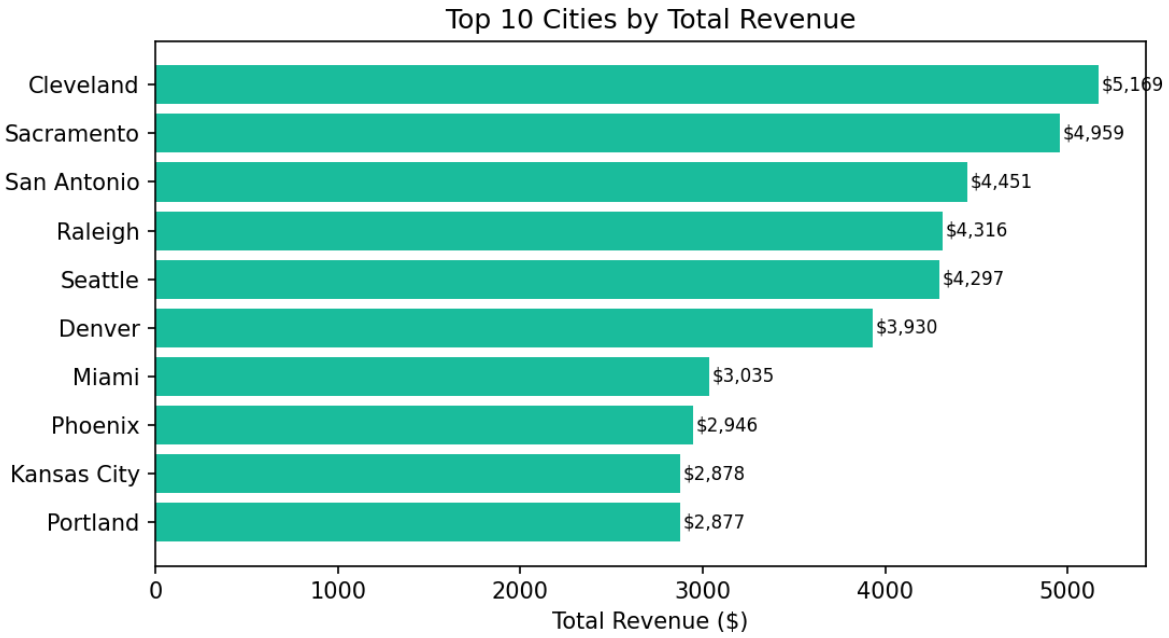
7.2 Satisfaction Rating by Payment Method



Debit Card orders show the highest average satisfaction at 4.14/5, while Credit Card orders sit slightly lower at 4.11/5 — the spread across payment methods is narrow, suggesting payment choice has little bearing on reported satisfaction.

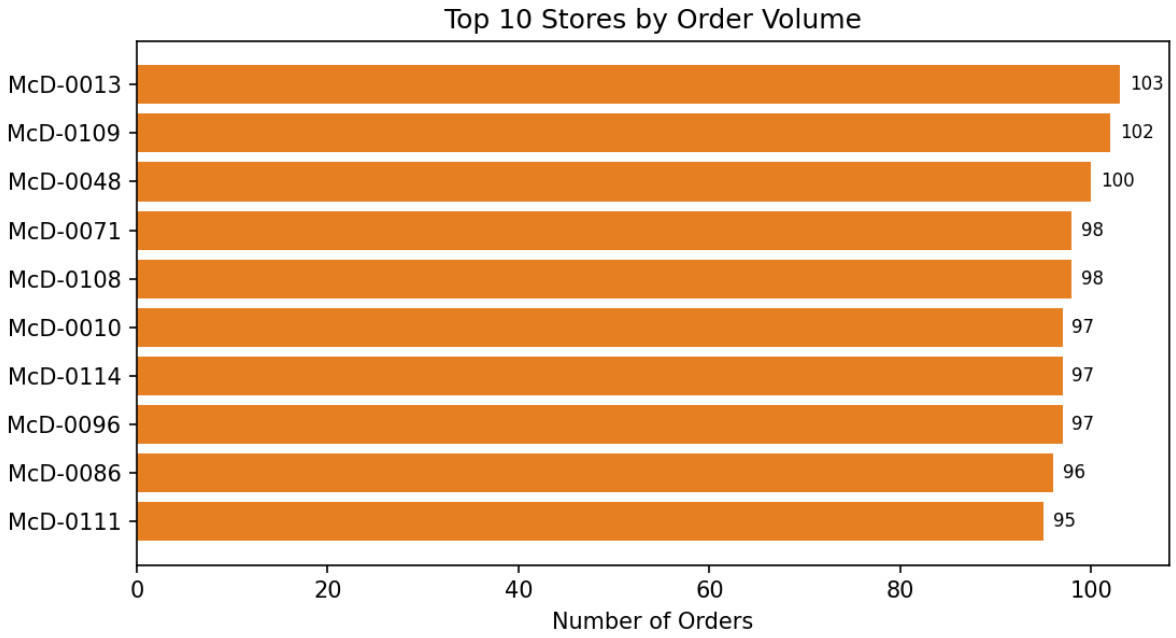
8. Store & City Performance

8.1 Top Cities by Total Revenue



Cleveland leads all cities with $5,169.47 in total net revenue, followed by Sacramento at $4,958.73 — revenue is fairly spread across the 20 cities in the dataset, with the top few cities somewhat ahead of the rest rather than perfectly even, reflecting the random store-to-city assignment used to build this dataset.

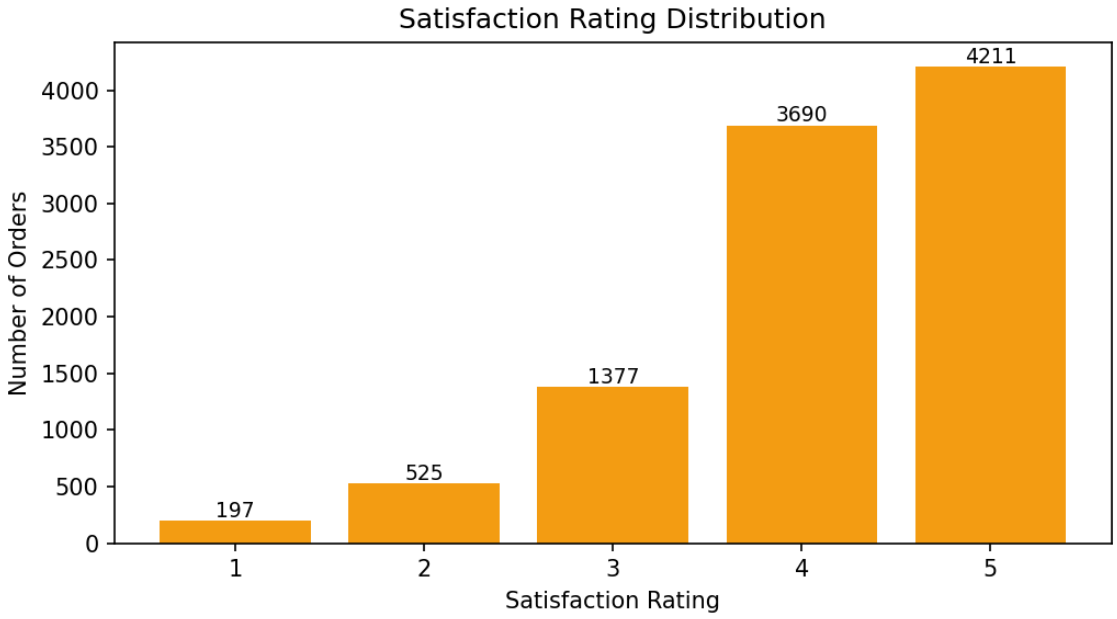
8.2 Top Stores by Order Volume



Store McD-0013 recorded the highest order volume at 103 orders across the record. With 120 stores sharing 10,000 total orders, individual store volumes cluster around the network average of roughly 83 orders per store.

9. Discount & Satisfaction Analysis

9.1 Satisfaction Rating Distribution



42.1% of orders received the top rating of 5, while only 7.2% received a low rating of 1 or 2 — the overall distribution skews strongly positive.

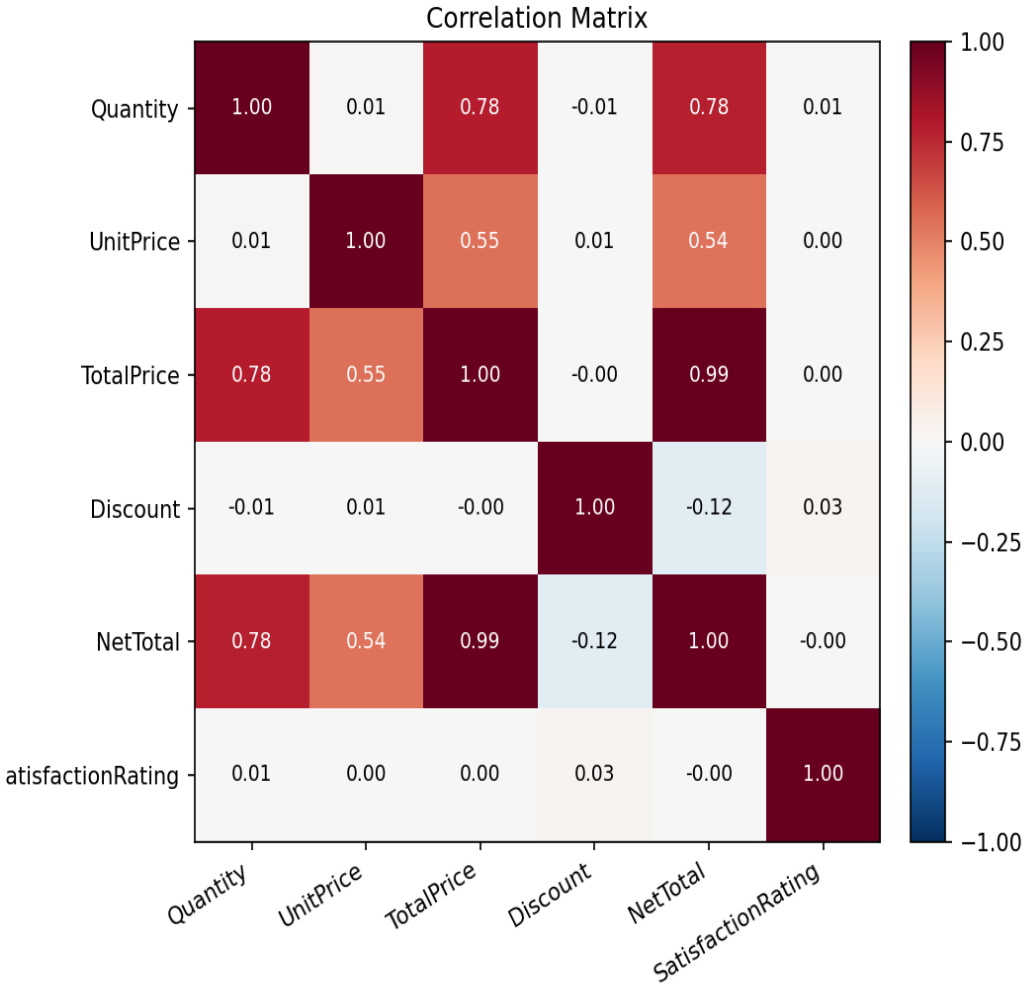
9.2 Discount Amount Distribution



Among the 1,112 orders that received a discount, the average discount applied was $1.49, with values spread across the $0.50–$2.50 range.

10. Correlation Analysis

10.1 Correlation Matrix



- Quantity and Total Price are strongly positively correlated (0.78): larger orders drive higher totals, as expected.
- Unit Price and Total Price show a moderate positive relationship (0.55), reflecting the mix of item price points across orders.
- Discount and Net Total show a mild negative relationship (-0.12) — discounted orders trend toward a lower final total, consistent with discounts being applied more to smaller baskets.
- Satisfaction shows essentially no relationship with quantity, price, or discount (all near 0.00–0.03), indicating none of these transaction attributes meaningfully predicts reported satisfaction.

10.2 What Drives Order Value

Order value is driven mainly by quantity and unit price rather than by discount level, order type, or payment method — the dataset shows no evidence that promotional discounting is correlated with either larger baskets or higher satisfaction.

11. Extreme Events

Event	Value	Order ID / Date
Highest single order	$33.20	ORD203232, 28 Feb 2025
Lowest single order (net)	$0.00	ORD200657, 16 Mar 2025
Largest quantity in one order	5 items	ORD200035 (Apple Pie)
Peak month (avg order value)	$5.98	Feb
Lowest month (avg order value)	$5.69	Jan

The record spans a range of $33.20 between the largest and smallest transactions, illustrating the full spread of basket sizes captured in this dataset. The single largest order, at $33.20, was a bulk purchase of 5 Quarter Pounders, consistent with a group or catering-style order.

Extreme transactions, while rare, matter disproportionately for planning: a single high-value order or a $0.00 net transaction can skew daily averages even though each represents a tiny fraction of total volume, which is why they are tracked separately from the monthly and yearly averages in Sections 4 and 9. In this dataset, 55 orders (0.55% of the record) net to $0.00 after a discount met or exceeded the item price.

12. Key Takeaways & Recommendations

• Order value is stable across the year: with monthly averages staying within a narrow band, revenue planning can treat seasonality as a minor factor compared to volume.

• Lunch-hour staffing matters most: with order volume peaking around midday, staffing and inventory should be weighted toward that window rather than spread evenly.

• Discounting shows no satisfaction lift: since discount level is uncorrelated with satisfaction, promotions should be evaluated on volume or traffic goals rather than expected satisfaction gains.

• Channel mix favors in-person and drive-thru: with Drive-Thru and Dine-In together accounting for over half of orders, operational investment should continue to prioritize those channels while still supporting digital growth.

• Store performance is broadly even, not concentrated: order volume per store clusters near the network average, so there is no single underperforming or overperforming outlier requiring urgent attention.

• Satisfaction is broadly positive: with 42% of ratings at the top score and under 8% at the bottom two, the current customer experience baseline is strong across payment methods and channels.

12.1 Priority Ranking

Priority	Recommendation	Impact Area
1	Weight staffing and inventory toward the midday order-volume peak	Operations, staffing
2	Evaluate promotions on volume/traffic goals, not satisfaction lift	Marketing, pricing
3	Continue prioritizing Drive-Thru and Dine-In while growing digital channels	Digital, in-store ops
4	Use store-level order volume as a baseline, not a red flag	Store management
5	Monitor extreme transactions ($0 net, bulk orders) separately from average	Finance, reporting

13. Next Steps: Descriptive → Predictive

This report answers what was recorded across 10,000 historical transactions. The natural next layer, if useful, moves from descriptive to predictive and prescriptive analytics:

- Short-term order-volume and revenue forecasting using the hourly and seasonal patterns identified here.
- Anomaly detection for extreme transactions (unusually large orders, $0 net orders) to trigger review alerts.
- Churn and repeat-visit prediction models using store, payment, and satisfaction data as leading features.
- Menu-mix optimization, if item-level cost data becomes available, to study margin alongside popularity.
- Live dashboard refresh — the same chart structure and section breakdown above would apply directly to a continuously updating point-of-sale feed.

13.1 How This Would Work at Real Scale

Layer	Purpose	Typical Tools
Ingestion	Capture each POS transaction in real time	Kafka / Kinesis from POS feed
Stream processing	Roll up raw orders into hourly/daily aggregates	Spark Structured Streaming / Flink
Storage	Hold raw transactions and pre-aggregated tables	Data lake (S3) + warehouse (Snowflake/BigQuery)
Batch aggregation	Nightly rollups: monthly averages, extremes, correlations	Airflow orchestrated SQL / dbt jobs
Serving / BI layer	Dashboards refreshed hourly for operations teams	Looker / Tableau / QuickSight

13.2 Data Enrichment Opportunities

Beyond the pipeline above, a few enrichments would meaningfully extend this report if the underlying feed supported them:

- Item-level cost and margin data to study profitability alongside popularity.
- Customer loyalty/membership IDs to distinguish repeat visitors from one-time orders.
- Weather data for each store's location, since foot traffic and drive-thru volume often correlate with conditions.
- Staffing levels per shift, to relate order-volume peaks to service speed and satisfaction.
- Marketing campaign flags, to attribute discount usage and order spikes to specific promotions.

14. Appendix: Data Tables

14.1 Monthly Averages

Month	Avg Net Total ($)	Orders
Jan	5.69	822
Feb	5.98	764
Mar	5.85	846
Apr	5.84	821
May	5.91	832
Jun	5.88	810
Jul	5.72	887
Aug	5.97	866
Sep	5.82	779
Oct	5.72	914
Nov	5.71	839
Dec	5.72	820

14.2 Yearly Averages

Year	Avg Net Total ($)	Orders
2024	5.94	5,053
2025	5.69	4,947

14.3 Top Menu Items by Order Count

Item	Orders
Apple Pie	935
Iced Coffee	913
Milkshake	739
Sprite (M)	647
Double Cheeseburger	607
Egg McMuffin	518
Sausage McMuffin	493
Orange Juice	459

14.4 Summary Statistics — All Numeric Fields

Metric	Mean	Std Dev	Min	25%	Median	75%	Max
Quantity	1.80	1.05	1.00	1.00	1.00	2.00	5.00
UnitPrice	3.31	1.34	1.19	2.02	3.03	4.09	7.03
TotalPrice	5.98	4.49	1.19	2.88	4.64	7.56	33.20
Discount	0.17	0.51	0.00	0.00	0.00	0.00	2.50
NetTotal	5.82	4.52	0.00	2.77	4.24	7.45	33.20
SatisfactionRating	4.12	0.97	1.00	4.00	4.00	5.00	5.00

15. Closing Notes

15.1 Methodology Recap

Monthly and yearly figures are simple means of Net Total across all orders within each grouping, computed from the parsed date field. Menu items are bucketed into three categories (Food, Beverage, Dessert) by item type. Correlations use Pearson's r across all numeric fields. The 55 orders that net to $0.00, where a discount met or exceeded the item price, are retained in all summary statistics for transparency.

15.2 Limitations

- This is a synthetic dataset generated for illustrative purposes — figures do not reflect actual McDonald's sales or operations.
- Orders are distributed across 120 stores and 20 cities using a random assignment, so geographic patterns here reflect sampling rather than real regional demand differences.
- No item-level cost data was available, so margin and profitability could not be assessed alongside popularity.
- Customer identity is not tracked, so repeat-visit and loyalty behavior could not be analyzed.

Ready when you are

Happy to build out any of the predictive extensions above, or re-run this entire report against a larger or live sales feed if one becomes available — the same chart structure and section breakdown above would apply directly.

Report generated from the McDonald's Sales Dataset (10,000 transaction records, Jan 2024 – Dec 2025). All figures are descriptive summaries of the provided synthetic sales data.